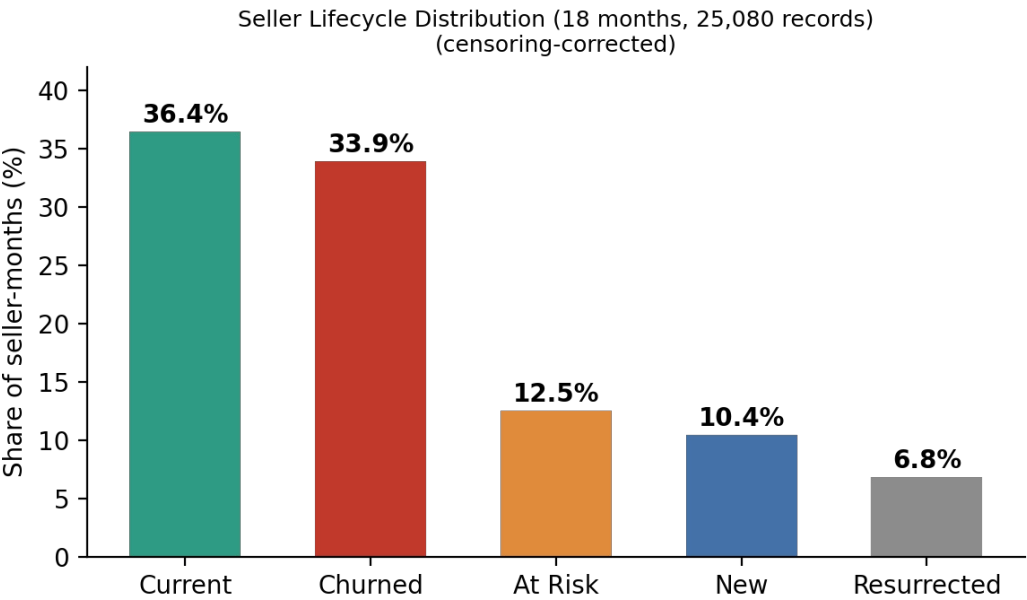


SELLERS 2,945 Olist marketplace	CHURNED SHARE 33.9% Of seller-month, corrected	TOP-SEGMENT GMV 69.6% From 14.9% of sellers	SEGMENTS FOUND 5 K-means (k=5)
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Research question: how can a marketplace convert one-time sellers into consistent, high-value sellers — and detect churn before it happens?

Seller Lifecycle Distribution

Headline Finding



Right-censoring-corrected: 33.9% of seller-month records are Churned, 36.4% are Current, and 12.5% are At Risk. Current — not Churned — is the largest category once records too recent to confirm are excluded.

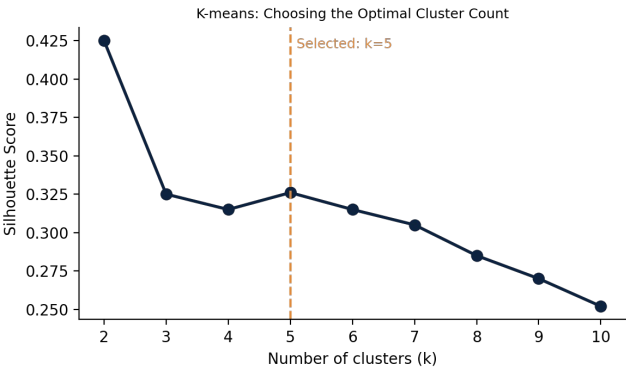
This is unevenly distributed — the Top Sellers segment (14.9% of sellers, 69.6% of GMV) is far more stable (median recency = 11 days), while the At-Risk segment already sits at 184 days of inactivity.

Confirmed by the Value Concentration and Churn Trajectory analyses on page 2. See page 3 for the before/after correction detail.

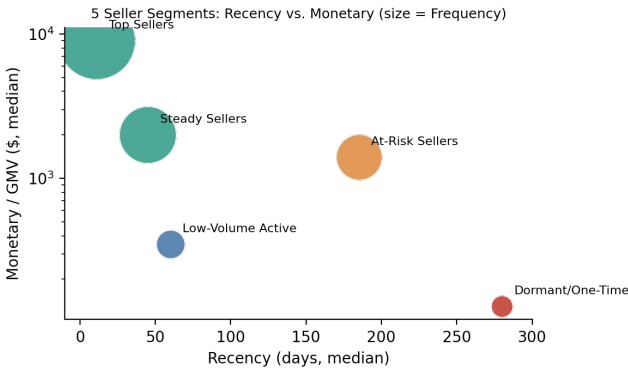
Methodology: Escalating Analysis Framework

1. Critical Event First Delivered Order — chosen for 96% coverage, confirmed via dataset audit.	2. Usage Interval Worksheet methodology: 80th-pct threshold = 42 days → mapped to Amplitude's Monthly interval (30-day operational threshold).	3. Lifecycle Framework New / Current / Resurrected / At Risk / Churned — 18 months (censoring-corrected), 25,080 records.	4. RFM + K-means Recency, Frequency, Monetary, Product Diversity → 5 segments (k=5, silhouette=0.326).
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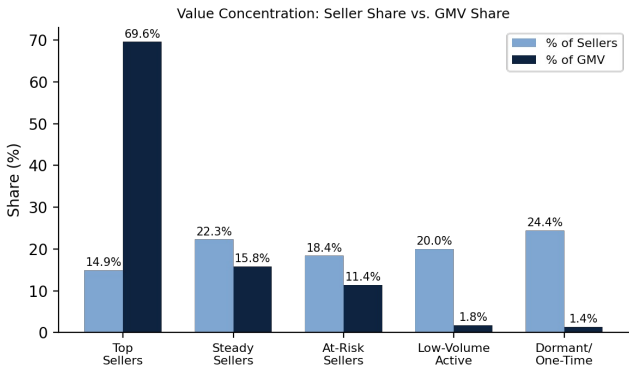
K-means: Optimal Cluster Count



5 Segments: Recency vs. Monetary



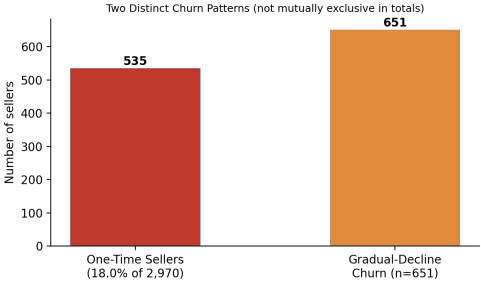
Value Concentration: Seller Share vs. GMV Share



Segment Table

Segment	% Sellers	% GMV
Top Sellers	14.9%	69.6%
Steady Sellers	22.3%	15.8%
At-Risk Sellers	18.4%	11.4%
Low-Volume Active	20.0%	1.8%
Dormant/One-Time	24.4%	1.4%

Two Distinct Churn Patterns

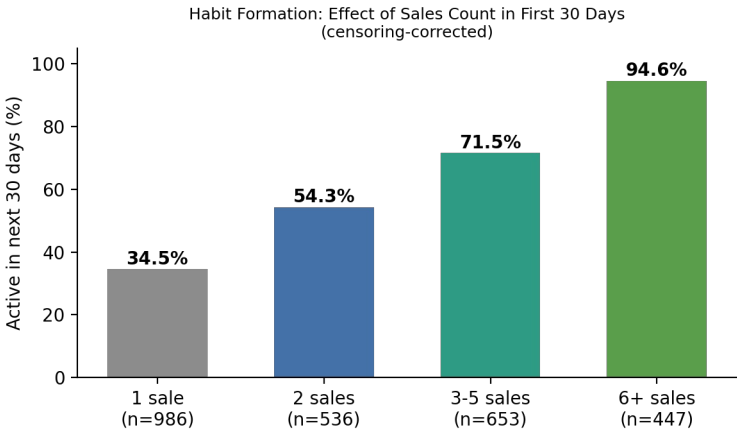


Decline % Before Churn (T-2 → T-0)

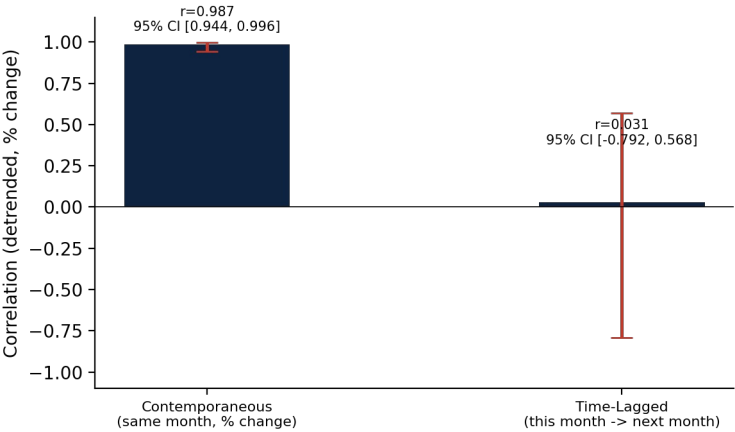
Metric	Decline
Orders	-35.6%
GMV	-41.2%
Product diversity	-24.7%

The 651-seller gradual-decline trajectory (right chart title) only covers sellers with 3+ active months. It misses the 535 one-time sellers (18%) who never returned after one sale — a structurally distinct churn pattern requiring a different fix (see page 3).

Habit Formation



Two-Sided Marketplace Dynamics



CORRECTED — All Three Flagged Issues Re-Run on Real Data (see Executive Summary for full detail)

Habit Formation right-censoring: excluding 323 of 2,945 sellers lacking a full 60-day window raised retention in every bucket (34.5% to 94.6%). Lifecycle Framework right-censoring: excluding the last 2 months (needed to confirm "Churned"), the distribution shifted from Churned 35.97%/Current 35.31% to Current 36.44%/Churned 33.91% — Current is now the largest category. Bootstrap 95% CI: same-month buyer-seller correlation confirmed real ($r=0.987$, CI [0.944, 0.996]); one-month-lagged correlation is genuinely inconclusive ($r=0.031$, CI [-0.792, 0.568], $n=18$ pairs) rather than "no relationship." No open methodological items remain.

Business Recommendation

Prioritize New Seller Activation and At-Risk Intervention: the former rests on the Habit Formation finding (30-day activity threshold), the latter on the Churn Trajectory finding (35-40% GMV-decline trigger point). Separately, address the 18% of sellers who never make a second sale with a dedicated first-to-second-sale nudge. Top Sellers (69.6% of GMV) warrant a high-touch retention strategy; Dormant/One-Time sellers warrant only a low-cost, automated approach.

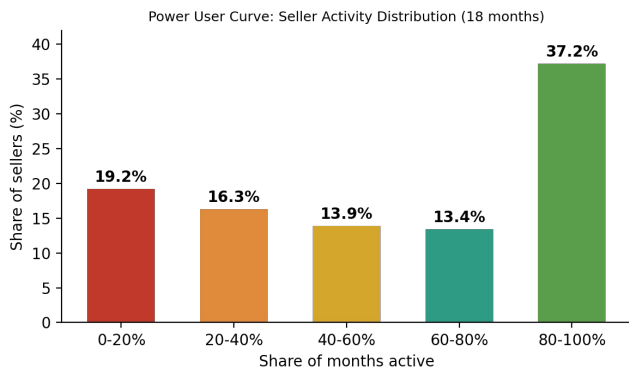
Limitations

- Pilot period (2016-09—12) excluded — 20-month raw window (18 after Lifecycle censoring correction).
- Two-Sided analysis is aggregation-level only (96.58% of buyers use 1 seller).
- Habit Formation / Churn Trajectory are correlational, not predictive.
- Findings specific to Olist (Brazil) — validate against local data before applying to Uzum Market.

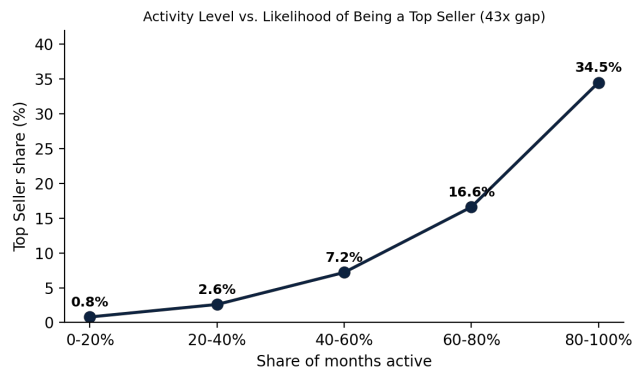
What Makes This Rigorous

- Pilot period identified and excluded via explicit statistical check.
- Critical event chosen by comparing 2 candidates on coverage.
- Right-censoring corrected in both Habit Formation and Lifecycle Framework.
- Bootstrap 95% CI added — revealed the lag correlation is inconclusive.
- K-means cluster count chosen via Elbow + Silhouette, not guessed.

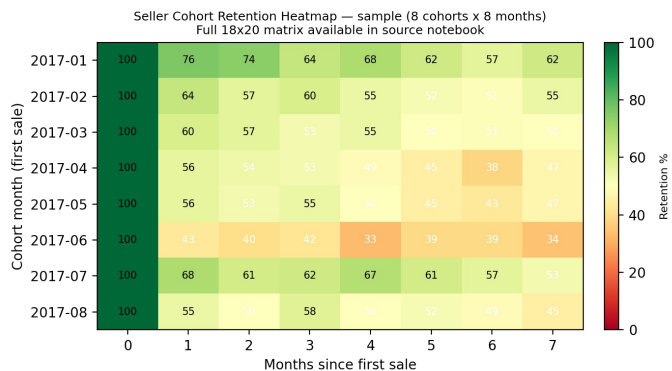
Power User Curve



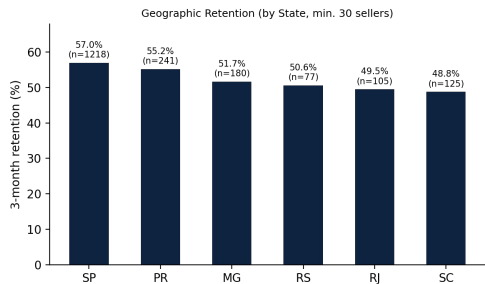
Activity Level Is Strongly Associated With Top-Seller Status



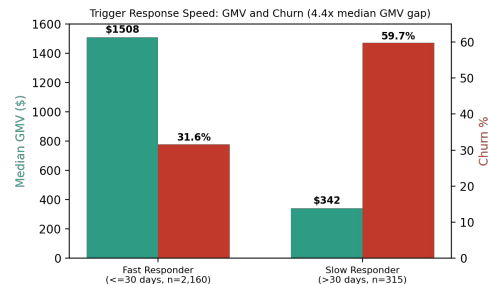
Seller Cohort Retention Heatmap (verified sample: 8 cohorts x 8 months)



Geographic Retention (negative finding)



Engagement Trigger Proxy



Among all 2,053 qualifying sellers, SP accounts for 59% (1,218); only states with 30+ qualifying sellers are shown, so the bars above don't sum to 2,053.

Fast Responders (≤ 30 days between sales, 87% of *measurable* sellers) have 4.4x higher median GMV (\$1,508 vs. \$342) — 6.5x on *average* GMV so median is the more reliable comparison — and about half the churn rate of Slow Responders.